Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Su Ryon Shin	Email	sshin4@bwh.harvard.edu
Institute Name	Mass General Brigham (Harvard Medical School)			
Address	65 Landsdowne Street, Rm. 281, Cambridge, MA 02139			

2. Technology Overview

Title of Proposed Technology	Handheld In Situ Electrospinning Platform for Chronic Wound Healing Using ECM homeostasis-Regulating Micromaterial–Embedded Nanofiber Matrices			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio & Medical		Bio & Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Pharmaceutical Biotechnology	Bio-based Biomaterials	Therapeutic and Diagnostic Devices	Other Therapeutic Devices
Concept of Proposed Technology	The proposed technology aims to develop a nanofiber-based wound dressing incorporating regenerative micromaterials that regulate extracellular matrix (ECM) homeostasis, thereby enabling localized and sustained bioactive signaling at the wound site and ultimately enhancing chronic wound healing. Regenerative micromaterials that contain ECM proteins derived from human stem cells can mimic the composition, morphological, biological, and physical properties of native ECMs, thereby enhancing ECM homeostasis. The multimodal nanofiber-based wound dressing with tunable biodegradation can prevent external infection while allowing oxygen to pass through and enabling localized, prolonged bioactive signaling at the wound site, thereby providing a regenerative environment. In addition, the technology is designed to form conformal nanofiber layers that adapt to irregular wound geometries across wide areas through on-site nanofiber deposition using a portable, handheld electrospinning approach. This enables the creation of customized, bioactive wound dressings directly on the wound surface for individual patients, overcoming the limitations of prefabricated dressings and supporting tissue regeneration.			
Rationale for Joint R&D Project	Regenerative biomaterials have emerged as key enabling components for next-generation wound care technologies due to their ability to stimulate cellular responses and support tissue regeneration. However, significant technical challenges remain in translating these biomaterials into medical device formats, including maintaining biological functionality, ensuring manufacturing reproducibility, and adapting to irregular wound geometries. Conventional pre-fabricated or sheet-type wound dressings are limited in their ability to accommodate			

	diverse wound shapes and depths encountered in real clinical settings. As a result, there is a growing need for advanced approaches that can effectively deliver the bioactive functions of regenerative biomaterials while providing conformal and customizable coverage of wound sites. This joint R&D project addresses these challenges by integrating regenerative biomaterial-based bioactive components into functional polymer nanofiber structures. Within this collaborative framework, an overseas research institution (MGB) contributes expertise in the design, mechanism investigation, and preliminary validation of stem cell-derived regenerative micromaterials, while Korean SMEs focus on multi-modal nanofiber-based wound dressing fabrication using medical-grade biodegradable polymers, process optimization with tunable biodegradation in depth of wounds, and translation into medical device applications via handheld in-situ electrospinning system which possess excellent adaptability to the medical field. Such an international collaborative approach is essential to bridge the gap between fundamental regenerative biomaterial research and clinically applicable wound dressing products. Through joint R&D, the project aims to enhance translational feasibility, accelerate product development, and establish a foundation for future entry into the global wound care market.
Key Words	1. Stem cell-derived regenerative micromaterials 2. Multi-modal nanofiber-based wound dressing 3. Localized and sustained delivery of bioactive signaling at the wound site 4. Handheld in-situ electrospinning system 5. Wound healing via enhanced ECM homeostasis

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The overall objective of this research is to establish ECM homeostasis-regulating micromaterial–Embedded Nanofiber Matrices that enable conformal application to irregular wound surfaces and shapes via a handheld in situ electrospinning platform.			
R&D Target Specifications	1. Enhancing ECM homeostasis of regenerative micromaterials and wound closure with minimal scar formation 2. Ensuring controllability of average nanofiber diameter and pore structure 3. Establish tunable biodegradation in depth (fast to slow degradation from the inside to outside of wounds) 4. Ensuring localized and sustained bioactive signaling 5. Establishment of stable and reproducible nanofiber formation suitable for clinical application environments 6. Conformal adhesion and applicability to irregular wound surfaces and depth 7. Satisfaction of basic physical properties and biocompatibility levels required for medical device applications			
R&D Plan	**Phase 1: Development and characterization of stem cell-derived regenerative micromaterials (0-12 months)** The ECM homeostasis of regenerative micromaterials and the conditions for maintaining biological functionality will be investigated. Biological responses with in vitro studies will be evaluated at the laboratory scale to determine suitability for wound healing. • Fabrication of human iPSC-laden stem cell fate-controlling microgels using a high-throughput droplet fabrication method • Differentiation of stem cell to fibroblasts that induced ECM deposition in the microgels, followed by decellularization to remove immunogenic cellular components (i.e., DNA) • Proteomic analysis to analyze secreted proteins and ECMs in the obtained microgels • In vitro studies to confirm biocompatibility, vascularization, dermal fibroblast migration, and ECM remodeling ability etc. **Phase 2: Fabrication and optimization of medical-grade polymer-based nanofibers (13-18 months)** Nanofiber fabrication conditions will be established using medical-grade polymers, including PUD, PEO, PVA, and HA. Formulations of two different polymers that achieve tunable biodegradation in depth will be designed, and process optimization will be conducted with a focus on fiber formation stability, structural characteristics, and reproducibility. • Develop multi-nozzle-based handheld in-situ electrospinning system for electrospinning two different medical-grade polymers, such as biodegradable and non-biodegradable, and make it easy to switch between the two polymers • Fabricate nanofiber-based wound dressing with tunable biodegradation in depth (fast to slow degradation from the inside to outside of wounds) • Implementation of nanofiber wound dressing structures adaptable to irregular wound geometries with advanced system **Phase 3: Development of bioactive nanofiber wound dressing structures (19-24 months)** Nanofiber structures capable of stably incorporating and retaining regenerative micromaterials will be developed. Conformal nanofiber coverage applicable to irregular wound geometries will be implemented, and structural adaptability and application feasibility will be evaluated. • Develop regenerative micromaterials-laden pre-polymer solutions for nanofiber fabrication while preserving biological functionality			

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	• Development of techniques for stable incorporation and uniform distribution of regenerative biomaterials within nanofiber structures • In vitro studies to confirm biocompatibility, vascularization, dermal fibroblast migration, and ECM remodeling ability etc. **Phase 4: Preliminary performance and applicability evaluation (25-36 months)** The basic therapeutic properties of the developed regenerative micromaterials-laden nanofiber-based wound dressing will be evaluated, along with preliminary biocompatibility, wound-healing function in a murine wound model, and basic efficacy assessments relevant to medical device applications. The potential for further development toward productization and clinical translation will also be reviewed. • Evaluate in vivo biocompatibility and biodegradation using a subcutaneous animal study • Evaluate in vivo wound healing ability using the murine wound defect model

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The overseas research institute will focus on the development and scientific validation of stem cell-derived regenerative micromaterials. This includes characterization of material properties, investigation of underlying mechanisms, and evaluation of biological functionality relevant to wound healing applications in vitro and in vivo. The institute will establish foundational technical knowledge required for nanofiber integration and support the definition of technical directions and performance goals during the early stages of the joint R&D project. In addition, the overseas research institute will identify conditions necessary to preserve bioactivity and conduct laboratory-scale validation to enhance the overall technical robustness of the collaborative research.
Expected R&R of Korean SME Partner	The Korean SME partner will be responsible for translating regenerative biomaterial-based bioactive components into functional polymer nanofiber-based wound dressing applications. This includes the establishment and optimization of nanofiber fabrication processes using medical-grade polymers such as PUD, PEO, PVA, and HA, as well as the development of techniques to ensure stable incorporation and reproducibility of bioactive components within nanofiber structures. The Korean SME will also implement nanofiber wound dressing configurations adaptable to irregular wound geometries and conduct basic physical property evaluation and biocompatibility assessment in consideration of medical device applications. Through these activities, the Korean SME partner will support the applicability of joint R&D outcomes and lay the groundwork for future product development and commercialization.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Through this joint R&D project, a core technology for integrating regenerative biomaterial-based bioactive components into functional polymer nanofiber structures is expected to be established, providing a technological foundation for bioactive nanofiber wound dressings adaptable to irregular wound sites. This approach is anticipated to overcome the limitations of conventional prefabricated or sheet-type dressings and introduce a next-generation wound-dressing technology that can be customized and applied directly at the point of care. From a technical perspective, the project is expected to enhance the practical applicability of regenerative biomaterials by achieving preservation of bioactivity, stable incorporation into nanofiber structures, and reproducible fabrication processes. The combination with functional polymer nanofiber technology is anticipated to improve the efficiency of bioactive signal delivery at wound sites and enable further expansion into a versatile platform technology for various wound care applications. From an industrial and commercial standpoint, the outcomes of this joint R&D project are expected to support the development of differentiated bioactive wound dressing products targeting the global wound care market. By strengthening technical maturity through international collaboration, the project will help establish a solid foundation for future medical device regulatory pathways and overseas market entry, thereby contributing to the global competitiveness of Korean SMEs and the expansion of international partnerships. Overall, this project is expected to generate both scientific and industrial impact by demonstrating an effective international joint R&D model that integrates regenerative biomaterial-based technologies with nanofiber manufacturing approaches, thereby contributing to technological innovation and global collaboration in advanced wound care.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization